EIGEN · COMPLEX PAIRS

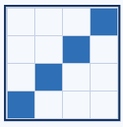
Three regimes, set by the modulus

The argument θ fixes how fast the plane turns and never affects whether it grows. That is decided entirely by the modulus, and the three cases below are the only ones there are.

3

REGIMES

NEITHER GROWING NOR DECAYING 1



On the unit circle | $r = 1$

GEOMETRIC ACTION

pure rotation by θ in the basis of the real canonical form, no scaling

DET OF THE 2×2 BLOCK

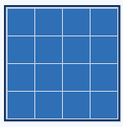
$\det A = 1$

ORBIT OF A POINT

a closed ellipse (a circle in the basis of the real canonical form)

In the basis of the real canonical form, lengths are preserved exactly and the block is **orthogonal**; A itself need not be. Every orbit closes if θ is a rational multiple of 2π and fills the circle densely otherwise — the boundary case in every stability argument.

GROWING OR DECAYING 2



Outside the unit circle | $r > 1$

GEOMETRIC ACTION

rotation with outward scaling by r

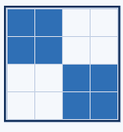
DET OF THE 2×2 BLOCK

$\det A > 1$

ORBIT OF A POINT

a spiral outward

Each application turns by θ and stretches by r, so iterates diverge while circling. In a discrete system this is instability — and the divergence is geometric, so it is fast.



Inside the unit circle | $r < 1$

GEOMETRIC ACTION

rotation with inward scaling by r

DET OF THE 2×2 BLOCK

$\det A < 1$

ORBIT OF A POINT

a spiral inward to the origin

The stable case for iteration: every orbit converges to the origin, still turning as it goes. This is what $|\lambda| < 1$ means geometrically, and why the condition is stated on the modulus rather than on the real part.

The determinant of the 2×2 block is $a^2 + b^2 = r^2$, so the **area scaling** and the modulus are the same fact stated twice — which is why the determinant can be read as a stability marker without computing an eigenvalue at all.